CSD TEAM – 12

HAND GESTURE DETECTION

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Hand Gesture Detection Project – Abstract

Hand gesture detection is a technology that recognizes and interprets human hand movements using a camera and software. This project aims to develop a system that detects and identifies different hand gestures in real-time. It uses computer vision techniques and machine learning models to analyze hand shapes, positions, and movements.

Technologies Used in Hand Gesture Detection

Hand gesture detection uses a combination of hardware and software technologies to recognize and interpret hand movements. Some of the key technologies include:

1. Computer Vision

- Helps analyze images and videos to detect hand shapes and movements.
- Uses techniques like **edge detection**, **contour detection**, and **feature extraction**.

2. Machine Learning & Deep Learning

- **Machine Learning (ML):** Uses trained models to recognize hand gestures based on patterns in data.
- **Deep Learning (DL):** Uses neural networks like **CNN (Convolutional Neural Networks)** to improve accuracy in detecting complex hand gestures.

3. Image Processing

- Uses libraries like **OpenCV** to process images and detect hands by applying filters, detecting edges, and removing background noise.

4. Sensor-Based Technology

- Uses **Infrared Sensors**, **Ultrasonic Sensors**, or **Radar Sensors** to track hand movements in the air.
- Example: **Leap Motion Controller** and **Microsoft Kinect** use sensors to detect 3D hand gestures.

5. Artificial Intelligence (AI)

- Helps the system learn and improve gesture recognition over time by analyzing large datasets.

6. Natural Language Processing (NLP) (For Sign Language Translation)

- If used for sign language recognition, NLP helps convert detected gestures into meaningful text or speech.

7. Hardware Components

- **Cameras (Webcam, Infrared Cameras, Depth Sensors):** Capture hand movements.
- **Microcontrollers (Arduino, Raspberry Pi):** Process gesture inputs for controlling devices.